

Devisree Sumesh

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EDUCATION

Amrita Vishwa Vidyapeetham

B.Tech, Electrical and Computer Science Engineering — Minor: Automation & Robotics — CGPA: 8.61

Coimbatore, India

2023 – Present

Chavara Public School

Class 12, CBSE — 91.7%

Pala, India

2021 – 2023

The Indian School

Class 10, CBSE — 97.4%

East Riffa, Bahrain

2009 – 2021

RESEARCH

AI in Robotic Surgery – Research Attachment, BDF Royal Medical Services, Bahrain

2026 – Ongoing

- Proposed and conducting independent IRB-approved research (RMS-MH/IRB/2026-1180) applying deep learning to Hugo RAS robotic surgical video, supervised by Dr. Ayman Raees (Robotic Surgery Programme Director, BDF Hospital) and facilitated by Prof. Martin Corbally (RCSI Bahrain).

μPAD Liver Enzyme Spectral Analysis – AMBE Lab, Amrita

2025 – Ongoing

- Contributing to supervisor-led project under Dr. P.E. Resmi: processing AS7341 11-channel spectral sensor data from paper microfluidic diagnostic strips for enzymatic colorimetric liver enzyme detection.

Hierarchical Safe DRL with Event-Triggered CBF for UAV Dynamic Obstacle Navigation

2026 – Ongoing

- Designing a hierarchical framework for single-UAV autonomous navigation combining Soft Actor-Critic (SAC) for path planning, Control Barrier Function with quadratic programming (CBF-QP) for formal collision safety guarantees, and an event-triggered replanning mechanism for computational efficiency.
- Implemented custom ObstacleAviary simulation environment in PyBullet .

Wearable Edge AI for Knee Rehabilitation Monitoring

2025

- first author; submitted to IEEE ISTAS Kerala 2026 (under review). Edge AI knee flexion classification (5 angle classes) on NodeMCU ESP8266 with MPU6050; quantised TFLite model via Edge Impulse; ThingSpeak telemetry; RehabVision 3D supervisory interface.

PROJECTS

Teleoperated FPV Robotic Arm with Surgical Vision Layer | [GitHub](#)

2025 – Ongoing

- * 4-DOF wireless teleoperation: potentiometer master arm → ESP32 ESP-NOW slave ESP32 → PCA9685 servo controller (MG996R/SG90 servos); ESP32-CAM wrist mount for live surgical field view.
- * YOLOv8n trained on CholecSeg8k and m2cai16 datasets (7 instrument classes, mAP50: 0.950); exported to ONNX for ROS 2 inference node integration.

ROS 2 Jazzy – Robot Simulation & Motion Planning | [GitHub](#)

2026

- * URDF modelling, TF2/RViz2 visualisation, Gazebo simulation; MoveIt2 MoveGroup interface for joint-space trajectory planning (OMPL RRTConnect, KDL kinematics).
- * Detection-to-motion pipeline: YOLOv8 label → joint constraint → MoveGroup goal.

ORGuard – Full-Stack Surgical Safety Platform | [Live](#) | [GitHub](#)

2026

- * MEN-stack (MongoDB/Express/Node.js) platform digitising the WHO Surgical Safety Checklist; REST API with 12 endpoints including aggregation pipelines for compliance analytics, pagination, and full-text search.
- * Chart.js dashboards, YOLOv8 surgical instrument inference page; WCAG AA accessible; deployed on Render with MongoDB Atlas.

Attention U-Net – Brain Tumour Segmentation | [GitHub](#)

2025

- * Trained Attention U-Net on LGG MRI Dataset (110 patients, 3929 slices); Dice: 0.8388, IoU: 0.7270 on held-out test set.
- * Attention gate mechanism with BCEDiceLoss; Grad-CAM on encoder activations for spatially interpretable tumour localisation heatmaps.

DC Motor PI Speed Control with Relay Auto-Tuning

2025

- * Closed-loop DC motor speed control on ESP32: custom IRFZ44N MOSFET H-bridge with IR2104 gate drivers; IR reflective sensor with IRAM_ATTR ISR for deterministic pulse counting.
- * Discrete PI controller ($K_p=0.897$, $K_i=0.267$) with anti-windup; on-device Åström–Hägglund relay-feedback auto-tuning eliminates offline plant identification. Written as IEEE-format project report.

TECHNICAL SKILLS

Languages: Python, C, SQL

ML / DL: PyTorch, TensorFlow/Keras, Scikit-learn, YOLOv8, OpenCV, MediaPipe, Edge Impulse

Robotics: ROS 2 (Jazzy), Gazebo, MoveIt2, URDF/Xacro, TF2, RViz2, PyBullet

Embedded: ESP32, STM32 (Cortex-M4), PIC16F877A, PCA9685, ESP-NOW, LoRa (SX1278), IR2104, IRFZ44N

Tools: Git, Google Colab, STM32CubeIDE, MPLAB X, LTSpice, MATLAB, L^AT_EX

AWARDS

Sastra Pratibha – Science India Forum (Vijnana Bharati), supported by Embassy of India, Bahrain & ISRO – 2015, 2018

LANGUAGES

English – Full professional proficiency